Compact Gaze-Only Reading-Goal Classification with Word-Aligned Teacher Distillation

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***Abstract*—Reading interfaces could adapt to whether a user reads for comprehension or seeks specific information. In prior work, the strongest single model combines gaze with contextual text and requires fixation-to-word alignment at inference. We use these resources only during training: a word-aligned teacher distills soft targets into a compact GRU that operates on fixation sequences alone. On OneStop, with both readers and articles held out, distillation improves balanced accuracy by 3.39 percentage points over the same GRU trained with hard labels. Reader-clustered and crossed reader/article bootstrap intervals exclude zero. Label smoothing does not reproduce the gain. Shuffling teacher logits within class reduces the improvement by 1.90 points, indicating that trial-specific targets contribute. Adding contextual text to the teacher yields no detectable student improvement. On a single-thread laptop CPU, model-only latency is 5.08 ms at the median and 16.57 ms at the 95th percentile.**

***Index Terms*—eye tracking, scanpath analysis, reading behavior, human–computer interaction, privileged information**

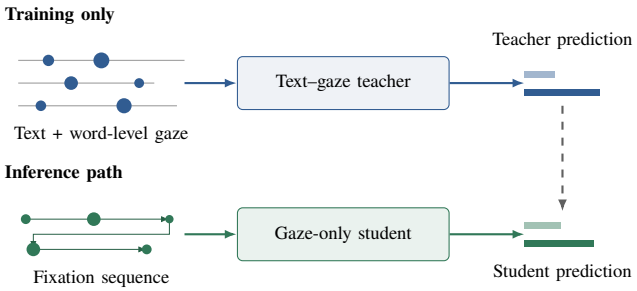


Fig. 1. Training-only distillation for gaze-only inference. A teacher predicts the reading goal from contextual text and word-level gaze summaries, and its soft predictions supervise a student that receives only the fixation sequence. The scanpath schematic is derived from an observed trial, with marker size representing fixation duration. At inference, the student operates without text, word alignment, or the teacher.

I. Introduction

A reader may approach the same passage to understand it broadly or to locate specific information. These goals are associated with differences in fixation durations, regressions, and scanpath structure [1], [2]. An opt-in reading-goal classifier could help reading interfaces offer navigation or content assistance after each paragraph [3]. Such a classifier must generalize to readers and articles absent from its training data.

Recent work established paragraph-level reading-goal decoding on OneStop Eye Movements and found that its strongest single model combined scanpaths with contextual language representations [3]. Its inference path requires passage text, fixation-to-word alignment, and a language encoder. We ask whether these resources can instead be confined to training while prediction uses gaze alone.

We study a text–gaze teacher that combines contextual word representations with word-level gaze summaries and a compact student that observes the chronological fixation sequence. The primary comparison holds the student architecture and input fixed. At inference, the student uses no teacher, text encoder, tokenization, or fixation-to-word alignment.

Each OneStop reader completes only one goal condition, while articles recur across readers. We therefore hold out readers and articles simultaneously and estimate uncertainty by resampling readers, with crossed reader/article resampling as a sensitivity analysis.

This study makes three contributions:

- With student inputs and architecture fixed, word-aligned teacher distillation improves the gaze-only GRU when both readers and articles are unseen.
- Controls using label smoothing, contextual-text removal, and target shuffling indicate that trial-specific soft supervision contributes even without contextual text.
- We characterize the student model's size and single-thread CPU latency across the observed sequence-length distribution.

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II. Related Work

A. Eye Movements and Reading Goals

OneStop includes eye movements from ordinary reading for comprehension, information seeking, and repeated reading [4]. It has also supported gaze-only and text–gaze models for predicting reading comprehension under new-reader and new-item conditions [5].

Shubi *et al.* introduced the reading-goal decoding task used here and evaluated gaze-only and multimodal models under new-reader and new-item conditions [3]. Their highest-performing single model retains text at inference. We instead hold the gaze-only inference architecture fixed and test whether richer training-time supervision improves it.

B. Distillation with Privileged Information

Knowledge distillation (KD) trains a student with class labels and a teacher's temperature-softened predictive distribution [6]. Generalized distillation allows the teacher to use privileged information that is unavailable to the student at inference [7]. This setting allows word-level text and gaze representations to guide training while the deployed model remains gaze-only. We apply it to reading-goal classification and test whether the benefit depends on contextual text or trial-specific teacher predictions.

III. Method

A. Task and Data

The unit of analysis is a completed paragraph-reading trial represented by its eye movements. The binary target distinguishes ordinary reading for comprehension ($y = 0$) from information seeking ($y = 1$). We use the anonymized public release of the OneStop Eye Movements corpus, collected with participant consent under an institutional ethics protocol [4]. We retain first readings from these two conditions and exclude repeated readings, question-screen events, answers, correctness, and all post-reading information.

The resulting data contain 19,438 trials from 360 readers and 30 articles, with nearly balanced goal labels. Each reader completed one goal condition, and the median trial contains 96 valid fixations. Reader identity, article identity, question content, session position, and condition metadata are not model inputs.

Each fixation is encoded by ten features. These are x and y coordinates normalized to the paragraph bounds, log fixation duration, horizontal and vertical displacement from the preceding fixation, log saccade amplitude, the sine and cosine of the saccade angle, a line-transition indicator, and position within the observed sequence. Retained fixations have positive finite durations and finite paragraph-screen coordinates. Continuous features are standardized using statistics from the current training set only.

B. Models

Table I summarizes the five neural model families. Models trained without distillation minimize binary cross-entropy (BCE). Gaze-summary MLP maps 22 sequence summaries to a 128–64 multilayer perceptron with ReLU and dropout 0.2. The summaries cover duration, displacement, saccade amplitude, line transitions, spatial dispersion, and path length.

TABLE I
Model inputs, objectives, and trainable parameter counts. The GRU architecture is held fixed.

Model	Objective	Model input	Parameters
Gaze-summary MLP	BCE	gaze summaries	11,265
Gaze GRU	BCE	fixation sequence	43,797
Text–gaze teacher	BCE	text + word-level gaze	1,038,945[a]
Text-only control	BCE	text	1,038,945[a]
Distilled GRU	BCE + KD	fixation sequence	43,797

[a] Trainable parameters; frozen RoBERTa is excluded.

Gaze GRU applies layer normalization to each fixation and processes the sequence with a two-layer unidirectional GRU [8] with hidden size 64. A 64–64–1 ReLU head with dropout 0.2 produces the logit. This model is the matched undistilled baseline for Distilled GRU. Sequences are zero-padded within each minibatch, and the classifier uses the GRU output at the trial's final fixation before padding.

A frozen RoBERTa-base encoder produces subword representations [9]. Text–gaze teacher averages the representations belonging to each paragraph word. Eight word-level gaze features encode fixation count, total and mean dwell time, first-fixation and first-pass dwell time, regression-in and revisit counts, and skipping. Text and gaze features are projected to 128 and 64 dimensions, concatenated with a 32-dimensional learned word-position embedding, and mapped to width 192. A learned classification token and a three-layer, six-head Transformer [10] with feed-forward width 384, GELU, and dropout 0.1 produce the logit. A padding mask excludes padded word positions. Text-only control retains the text branch and fusion architecture but sets all gaze features to zero. It tests whether paragraph text alone predicts the reading goal.

Distilled GRU has exactly the same architecture and inputs as Gaze GRU. Let z_S and z_T be the student and teacher logits, and let $\tau = 2$. The student minimizes

$$\mathcal{L}_D = (1-\alpha)\,\mathrm{BCE}(y, \sigma(z_S)) + \alpha\tau^2\,\mathrm{KL}\left[q_T^{(\tau)} \,\middle\|\, q_S^{(\tau)}\right], \quad q_j^{(\tau)} = \mathrm{Bern}\left(\sigma\left(\frac{z_j}{\tau}\right)\right). \tag{1}$$

Teacher logits are generated only for student-training trials within each outer split. No held-out test trial is used to fit either model or construct a distillation target.

C. Evaluation Protocol

Readers are divided into five goal-stratified folds, and articles are divided into five folds. Their Cartesian product gives 25 disjoint test partitions that together cover all retained trials once. For each reader–article fold pair, training excludes every trial from either held-out fold, and the test set is their intersection. All test readers and articles are therefore absent from training.

Within each outer training set, model selection uses three paired inner validation splits. Each validation split likewise holds out both its readers and articles from the corresponding inner training set. Learning rates are selected from $\{10^{-3}, 3\times 10^{-4}\}$ for the compact models and $\{3\times 10^{-4}, 10^{-4}\}$ for the teacher models. We also select $\alpha \in \{0.25, 0.50, 0.75\}$ for Distilled GRU. Balanced accuracy is the selection metric, with negative log-likelihood (NLL) as the first tie-breaker. Each final model is refitted for the median number of epochs selected across the three inner splits. The decision threshold is fixed at 0.5 before test evaluation. Test labels are not used for early stopping, calibration, or threshold selection.

Models are optimized with AdamW [11], weight decay 10^{-4}, and gradient clipping at 1.0. Inner training uses at most 30 epochs with a patience of five epochs. Batch sizes are 512 for Gaze-summary MLP, 256 for the GRU students, and 128 for the teacher models. Final refits use the selected learning rate and fixed epoch count described above, without test-dependent stopping.

D. Post Hoc Ablations

After the primary result, we ran post hoc controls using the same outer splits and training setup. Paired conditions inherit the selected learning rate and epoch count and keep the remaining student, teacher, or KD settings fixed.

The first control replaces teacher targets with binary label smoothing ($\epsilon = 0.10$) [12]. The second zeros every RoBERTa representation in an architecture-matched teacher while retaining its word-level gaze summaries. Across three random seeds, we refit the BCE and label-smoothed GRUs, both teachers, and both KD students. Each student uses logits from the teacher trained with the same seed.

The third control cyclically reassigns teacher logits within each ground-truth class, preserving the class-conditional distributions while removing trial-level correspondence. It uses the same seeds and student settings.

We also fit logistic controls using either log dwell time per word [3] or the 22 gaze summaries. Their regularization is selected within the same nested reader/article splits.

E. Metrics and Uncertainty

Balanced accuracy (BA) is the primary metric because it averages the two goal-specific recalls [13]. Secondary metrics are AUROC, macro F1, negative log-likelihood (NLL), Brier score, and 15-bin expected calibration error (ECE) [14].

For the primary contrast between Distilled GRU and Gaze GRU, a paired 5,000-draw percentile bootstrap resamples whole readers within each goal. We report the central 95% interval. A 2,000-draw sensitivity analysis resamples readers within each goal and independently resamples articles, weighting each trial by the product of their multiplicities. Bootstrap samples are paired between models. For each post hoc contrast, resampled units are paired across conditions, and the seed-specific BA differences are averaged within each draw.

TABLE II

Out-of-fold test performance with both readers and articles held out. Lower is better for NLL, Brier, and ECE. LR controls were added post hoc.

Model	BA	AUROC	Macro F1	NLL	Brier	ECE
Dwell-time LR[b]	.5887	.6292	.5887	.6635	.2359	.0155
Gaze-summary LR[b]	.6066	.6493	.6065	.6572	.2327	.0236
Gaze-summary MLP	.5942	.6359	.5936	.6632	.2360	.0410
Gaze GRU	.5866	.6178	.5862	.6710	.2391	.0210
Text–gaze teacher	.6126	.6613	.6120	.6561	.2312	.0496
Text-only control	.4986	.5017	.4986	.6952	.2510	.0304
Distilled GRU	**.6205**	**.6708**	**.6204**	**.6427**	**.2265**	**.0142**

[b]LR controls use nested selection within the same reader/article splits.

TABLE III

Matched-seed post hoc controls. For model rows, ΔBA is relative to GRU+BCE. Rows below the rule are direct contrasts. CIs are reader-clustered, and n_+ counts positive seeds.

Condition	BA	ΔBA	Reader 95% CI	n_+
GRU+BCE	.5857	–	–	–
+ label smoothing	.5852	−.0005	[−.0022,.0011]	1/3
+ gaze-only KD	.6221	+.0364	[.0199,.0534]	3/3
+ text–gaze KD	.6232	+.0376	[.0223,.0529]	3/3
+ class-shuffled KD	.6042	+.0185	[.0052,.0317]	3/3
Text–gaze KD − gaze-only KD	–	+.0011	[−.0030,.0052]	1/3
Text–gaze KD − class-shuffled	–	+.0190	[.0061,.0317]	3/3

F. CPU Measurement

We benchmark one distilled checkpoint on every retained fixation sequence. After 100 warm-up iterations, we time one full-sequence forward pass per trial in randomized order. Measurements use an Intel Core Ultra 7 155H and one PyTorch thread, excluding input construction, parsing, and disk I/O.

IV. Results

A. Confirmatory Performance

Table II reports pooled out-of-fold predictions for test readers and articles excluded from training.

Distilled GRU improves BA over Gaze GRU by 0.0339. The reader-clustered 95% interval is $[0.0175, 0.0504]$, and the crossed reader/article interval is $[0.0130, 0.0553]$. Its NLL, Brier score, and ECE are also lower than those of the matched GRU. The distilled model exceeds Gaze-summary MLP and Text–gaze teacher in pooled BA, while Text-only control remains at chance.

The strongest post hoc logistic control, Gaze-summary LR, reaches .6066 BA. The distilled model leads it by .0139 BA, with a crossed reader/article interval of $[-0.0037, 0.0321]$.

B. Post Hoc Ablations and Seed Sensitivity

Table III averages the post hoc comparisons over three matched random seeds. Label smoothing does not improve BA over the GRU. Students trained with either teacher, as well as class-shuffled KD, improve BA at every seed, with reader-clustered intervals above zero.

Text–gaze and gaze-only KD differ by .0011 BA, with both reader-clustered and crossed intervals spanning zero. Text–gaze KD exceeds class-shuffled KD by .0190 BA, with

a crossed interval of $[.0033, .0352]$. Class-shuffled KD still exceeds the GRU by .0185 BA.

C. Inference Cost

The distilled and undistilled GRUs have identical parameter counts and inference operations. The serialized student state occupies 176 KiB. Across all retained sequences, model-only latency has a 5.08-ms median and a 16.57-ms 95th percentile.

V. Discussion

Teacher-derived soft targets improve the fixed gaze-only GRU, whereas label smoothing does not. The gaze-summary logistic baseline remains competitive. The evidence therefore supports distillation as a better training objective for this sequence architecture.

Adding contextual text to the teacher yields no detectable student improvement. Both teachers use word-level gaze aggregation and have greater capacity than the student, leaving the contribution of these shared properties unresolved.

Within-class shuffling cuts the matched-seed gain roughly in half but does not remove it. Trial-specific teacher predictions account for part of the gain, while the remainder is consistent with class-dependent soft targets or their optimization effect.

The post hoc ablations were designed after the primary analysis and average three seeds. Their intervals therefore characterize reader and article sampling more directly than variability across initializations. Because reading goal is assigned between readers, a crossover study is needed to identify within-reader effects. The data come from one English news corpus and a laboratory eye tracker, and all predictions use completed scanpaths. Other domains, trackers, self-selected goals, and partial sequences require separate evaluation.

The gaze-only student still requires calibrated coordinates and paragraph bounds. End-to-end device studies should include eye tracking, preprocessing, memory, and energy, which are outside the CPU benchmark. Any interface adaptation should be opt-in and evaluated with users.

VI. Conclusion

Training-only distillation improves a compact gaze-only GRU when both readers and articles are unseen. The deployed model uses only fixation sequences. Within-class target shuffling indicates that trial-specific teacher predictions explain part of the gain, while contextual text adds no detectable benefit.

Code Availability

Code, configurations, and aggregate results are available at github.com/gohyunsu/gaze-goal.

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